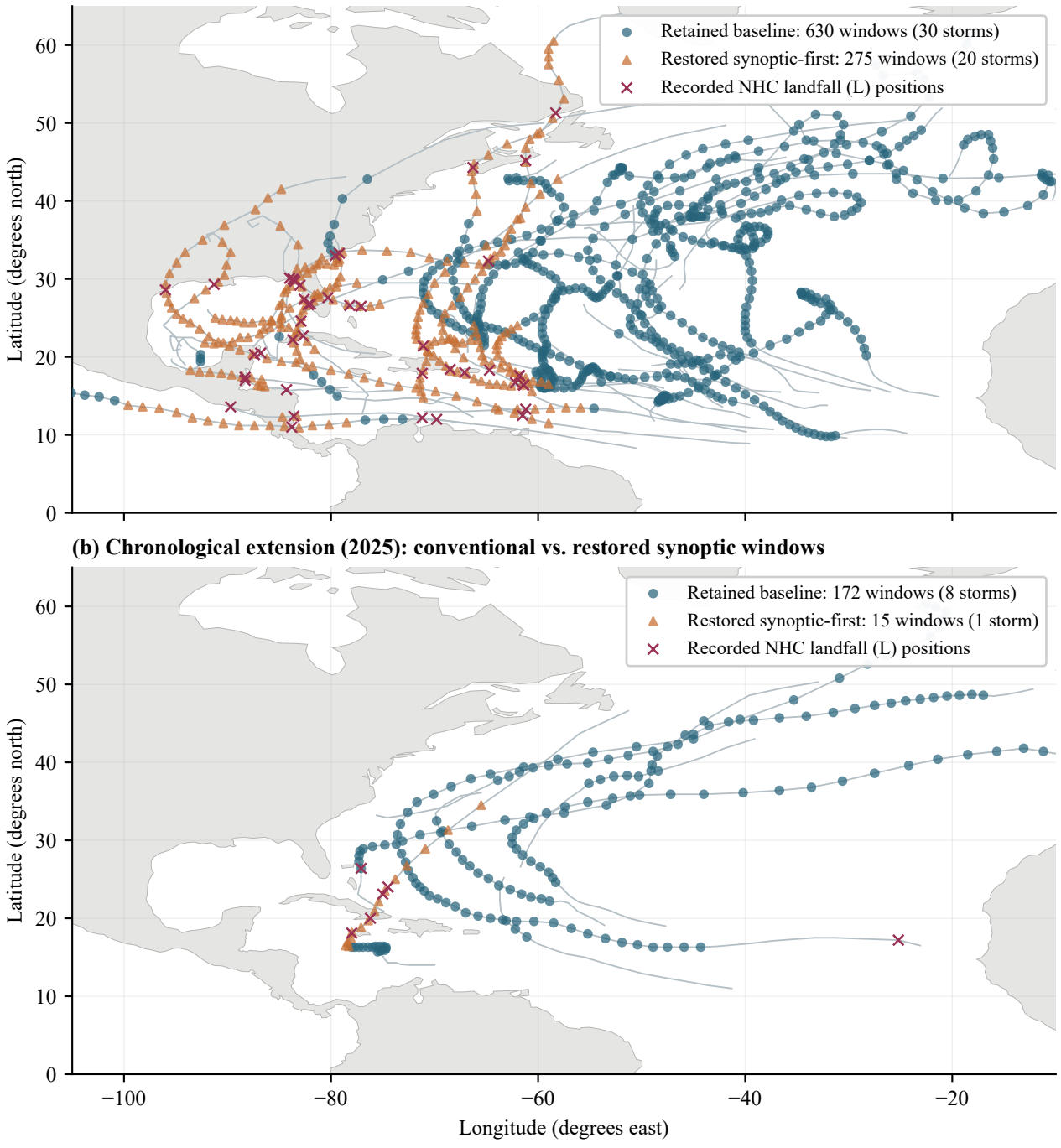

(a) Primary test cohort (2022–2024): conventional vs. restored synoptic windows
Retained baseline: 630 windows (30 storms)
Restored synoptic-first: 275 windows (20 storms)
Recorded NHC landfall (L) positions
Latitude (degrees north)
0
10
20
30
40
50
60
(b) Chronological extension (2025): conventional vs. restored synoptic windows
Retained baseline: 172 windows (8 storms)
Restored synoptic-first: 15 windows (1 storm)
Recorded NHC landfall (L) positions
Latitude (degrees north)
0
10
20
30
40
50
60
−100
−80
−60
−40
−20
Longitude (degrees east)